Formation of Bus Admittance Matrix

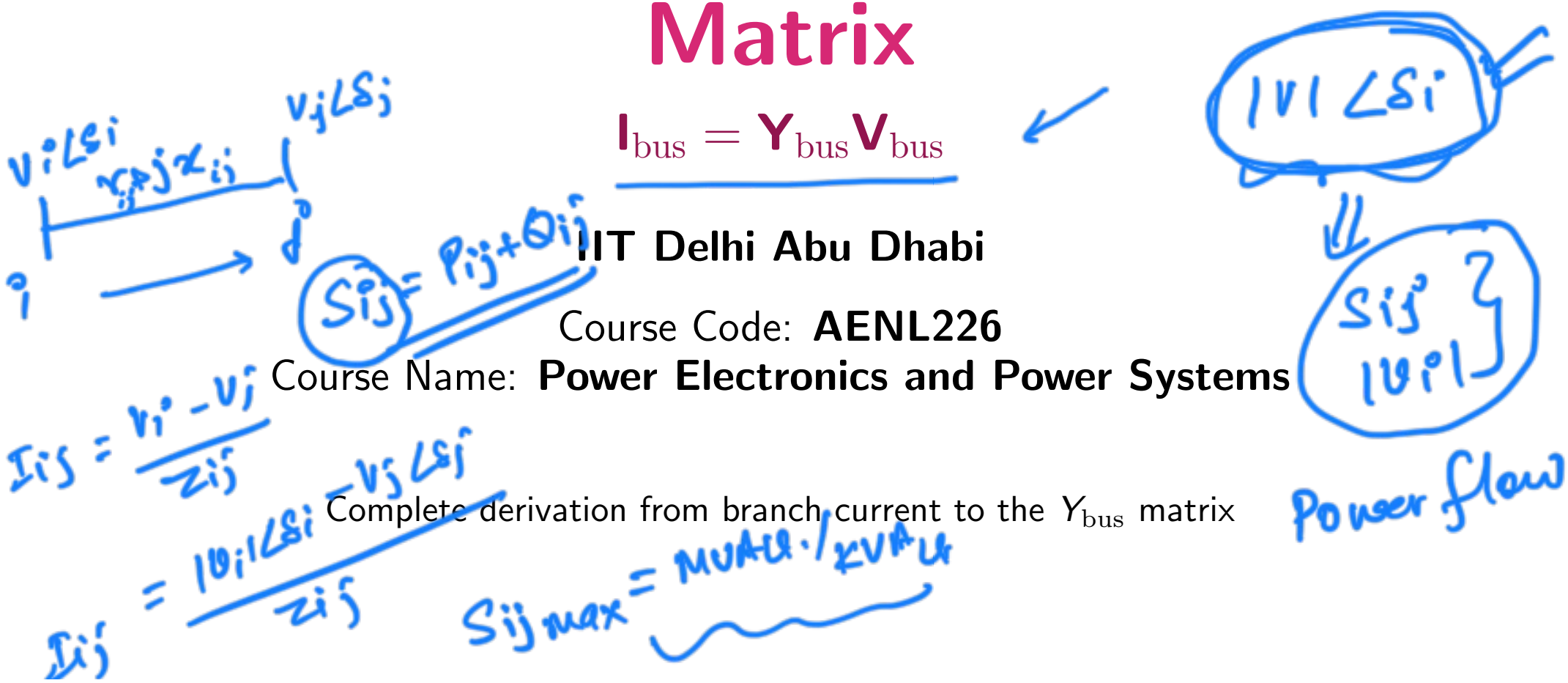
$$\mathbf{I}_{\text{bus}} = \mathbf{Y}_{\text{bus}}\mathbf{V}_{\text{bus}}$$

IIT Delhi Abu Dhabi

Course Code: **AENL226**

Course Name: **Power Electronics and Power Systems**

Complete derivation from branch current to the Y_{bus} matrix



Learning Path

We will derive the nodal current equation used in power-system analysis:

$$\boxed{\mathbf{I}_{\text{bus}} = \mathbf{Y}_{\text{bus}}\mathbf{V}_{\text{bus}}}$$

Classroom sequence

1. Start with one branch between bus i and bus j.
2. Derive the branch current I_{ij}.
3. Apply KCL at a general bus i.
4. Identify Y_{ii} and Y_{ij}.
5. Construct Y_{bus} for a 3-bus example.

Bus Voltages and Injected Currents

At every bus, voltage is measured with respect to the reference node:

$$V_1, V_2, \ldots, V_n.$$

The current injected into each bus is:

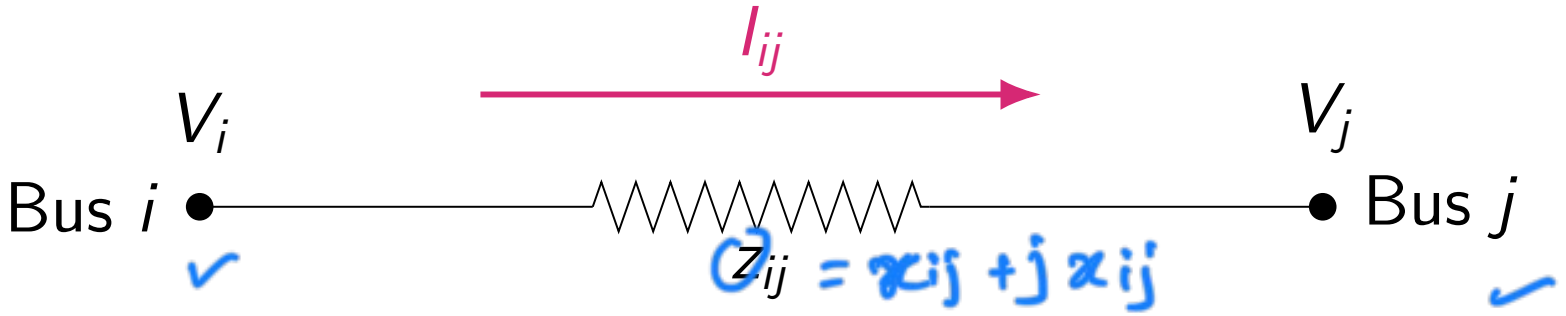
$$I_1, I_2, \ldots, I_n.$$

$$\mathbf{I}_{\text{bus}} = \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{bmatrix}, \qquad \mathbf{V}_{\text{bus}} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix}.$$

Sign convention

Positive I_i means current is injected into bus i from an external source.

One Branch Between Bus i and Bus j



The series admittance of the branch is

$$\boxed{y_{ij} = \frac{1}{z_{ij}}}.$$

Branch Current Derivation

The voltage drop from bus i to bus j is

$$V_i - V_j.$$

By Ohm's law,

$$I_{ij} = \frac{V_i - V_j}{z_{ij}}.$$

Since $y_{ij} = 1/z_{ij}$,

$$\boxed{I_{ij} = y_{ij}(V_i - V_j)}$$

Expanding,

$$\boxed{I_{ij} = y_{ij} V_i - y_{ij} V_j.}$$

Key point

The same line gives a positive coefficient for V_i and a negative coefficient for V_j in the bus-i equation.

Current in the Opposite Direction

If current is measured from bus j to bus i,

$$I_{ji} = y_{ij}(V_j - V_i).$$

Expanding,

$$I_{ji} = y_{ij}V_j - y_{ij}V_i.$$

Therefore,

$$\boxed{I_{ji} = -I_{ij}.}$$

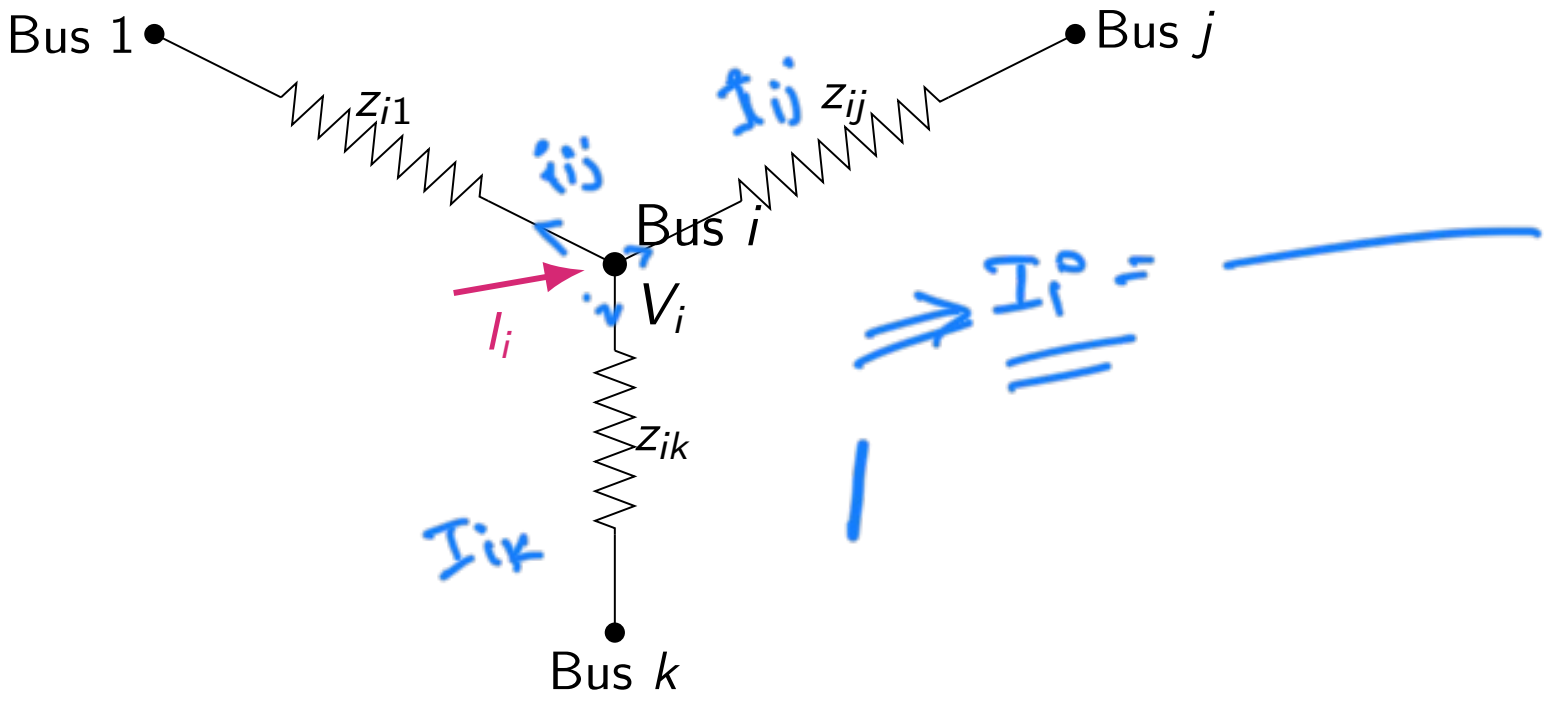
For a reciprocal line,

$$\boxed{y_{ij} = y_{ji}.}$$

This is why the final Y_{bus} matrix is usually symmetric:

$$\boxed{Y_{ij} = Y_{ji}.}$$

General Bus i Connected to Several Buses



Let $\mathcal{N}_i$ be the set of buses directly connected to bus i.

KCL at Bus i: Without Shunt

Using Kirchhoff's Current Law at bus i,

$$I_i = \sum_{j \in \mathcal{N}_i} I_{ij}.$$

For each connected branch,

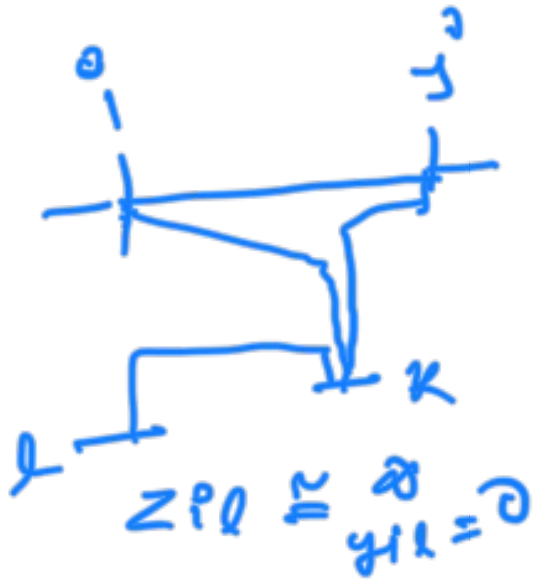
$$I_{ij} = y_{ij}(V_i - V_j).$$

Substituting,

$$I_i = \sum_{j \in \mathcal{N}_i} y_{ij}(V_i - V_j).$$

Expanding,

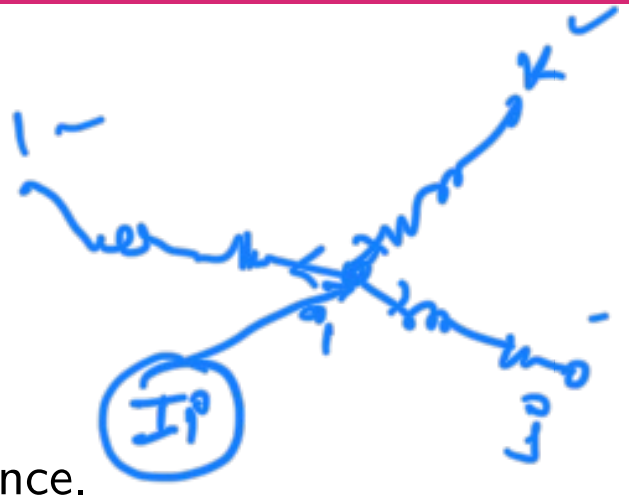
$$I_i = \sum_{j \in \mathcal{N}_i} y_{ij} V_i - \sum_{j \in \mathcal{N}_i} y_{ij} V_j.$$



Bus i Equation: Without Shunt

Since V_i is common in the first summation,

$$I_i = \left(\sum_{j \in \mathcal{N}_i} y_{ij} \right) V_i - \sum_{j \in \mathcal{N}_i} y_{ij} V_j.$$



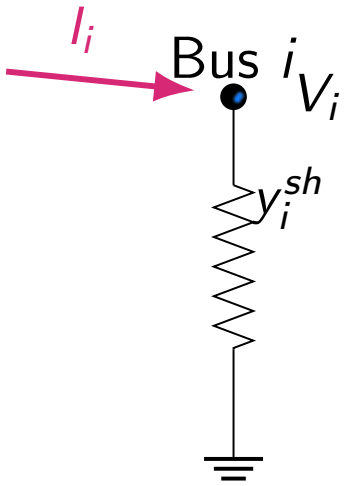
This is the bus-current equation for bus i without shunt admittance.

Coefficient meaning

$$\text{coefficient of } V_i = \sum_{j \in \mathcal{N}_i} y_{ij}$$

$$\text{coefficient of } V_j = -y_{ij}.$$

Adding Shunt Admittance at Bus i



The shunt current is

$$\boxed{I_i^{sh} = y_i^{sh} V_i.}$$

Therefore, shunt admittance only changes the coefficient of V_i.

KCL at Bus i: With Shunt

KCL at bus i becomes

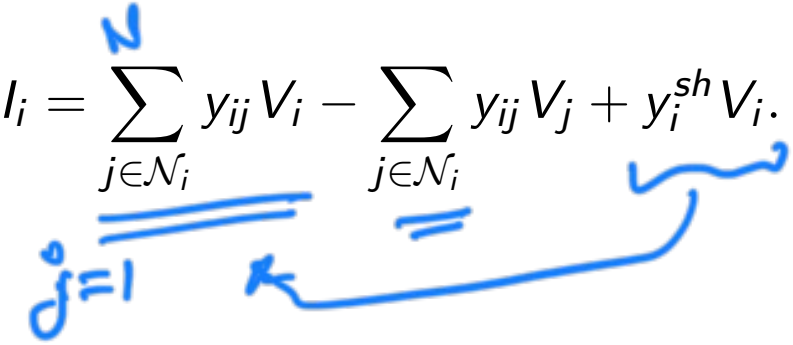
$$I_i = \sum_{j \in \mathcal{N}_i} I_{ij} + I_i^{sh}.$$

Substitute the branch and shunt currents:

$$I_i = \sum_{j \in \mathcal{N}_i} y_{ij}(V_i - V_j) + y_i^{sh} V_i.$$

Expanding,

$$I_i = \sum_{j \in \mathcal{N}_i} y_{ij} V_i - \sum_{j \in \mathcal{N}_i} y_{ij} V_j + y_i^{sh} V_i.$$



Final Bus i Current Equation

Collect the terms containing V_i:

$$I_i = \left(y_i^{sh} + \sum_{j \in \mathcal{N}_i} y_{ij} \right) V_i - \sum_{j \in \mathcal{N}_i} y_{ij} V_j$$

This is the complete current equation at bus i.

It can also be written as

$$I_i = Y_{i1} V_1 + Y_{i2} V_2 + \cdots + Y_{ii} V_i + \cdots + Y_{in} V_n$$

or compactly,

$$I_i = \sum_{j=1}^{n} Y_{ij} V_j.$$

Identifying Y_{ii} and Y_{ij}

Compare the bus-i equation

$$I_i = \left(y_i^{sh} + \sum_{k \in \mathcal{N}_i} y_{ik} \right) V_i - \sum_{j \in \mathcal{N}_i} y_{ij} V_j$$

with the standard row form

$$I_i = \sum_{j=1}^{n} Y_{ij} V_j .$$

Diagonal element

$$\boxed{Y_{ii} = y_i^{sh} + \sum_{k \in \mathcal{N}_i} y_{ik}}$$

Off-diagonal element

For $i \neq j$:

$$\boxed{Y_{ij} = -y_{ij}}$$

if connected; otherwise $\boxed{Y_{ij} = 0}$.

Direct Construction Rules

For a line between buses i and j with admittance y_{ij}:

$$\boxed{Y_{ii} \leftarrow Y_{ii} + y_{ij}}$$

$$\boxed{Y_{jj} \leftarrow Y_{jj} + y_{ij}}$$

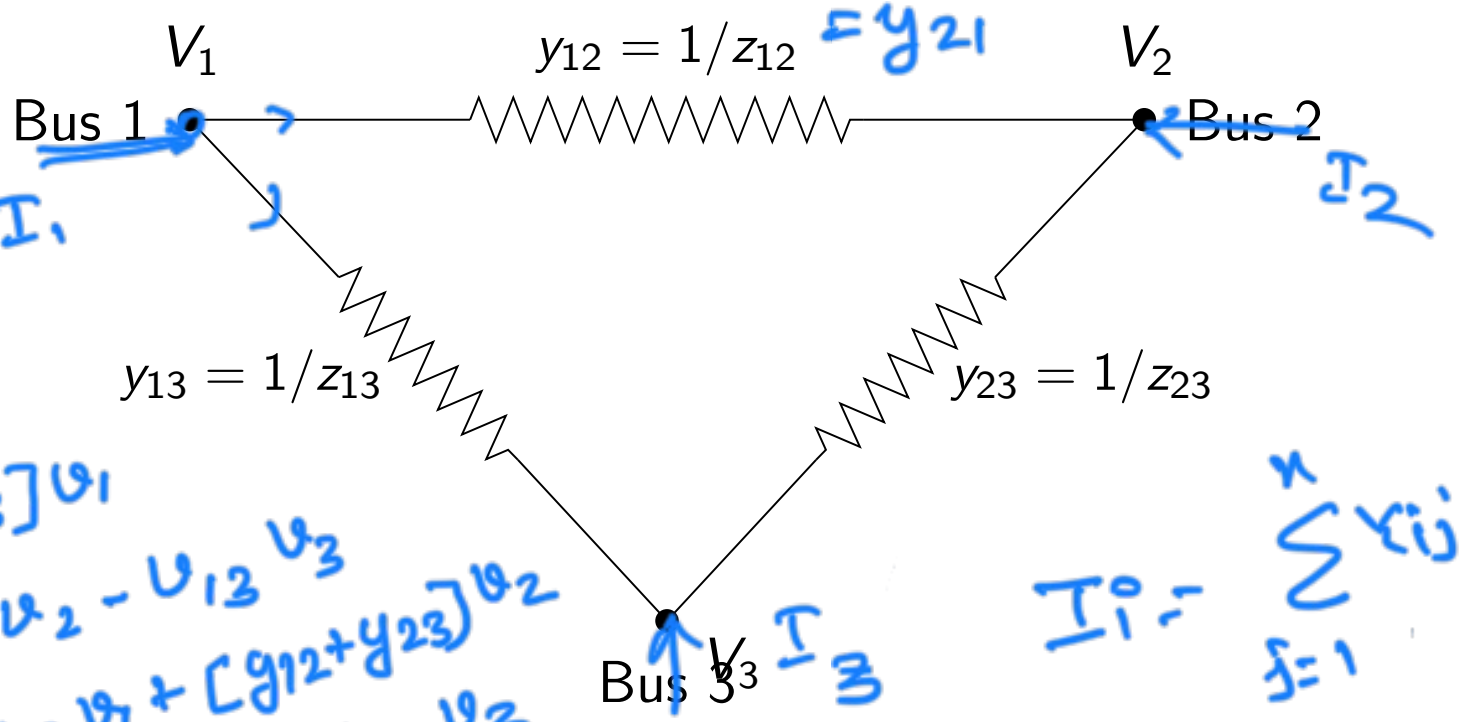
$$\boxed{Y_{ij} \leftarrow Y_{ij} - y_{ij}}$$

$$\boxed{Y_{ji} \leftarrow Y_{ji} - y_{ij}}$$

If a shunt admittance y_i^{sh} is connected at bus i,

$$\boxed{Y_{ii} \leftarrow Y_{ii} + y_i^{sh}.}$$

Three-Bus Network



Series admittances: y_{12}, y_{13}, and y_{23}.

KCL at Bus 1

At bus 1,

$$I_1 = I_{12} + I_{13}.$$

Substitute branch currents:

$$I_1 = y_{12}(V_1 - V_2) + y_{13}(V_1 - V_3).$$

Expand:

$$I_1 = y_{12}V_1 - y_{12}V_2 + y_{13}V_1 - y_{13}V_3.$$

Collect terms:

$$\boxed{I_1 = (y_{12} + y_{13})V_1 - y_{12}V_2 - y_{13}V_3.}$$

Thus,

$$Y_{11} = y_{12} + y_{13}, \quad Y_{12} = -y_{12}, \quad Y_{13} = -y_{13}.$$

KCL at Bus 2

At bus 2,

$$I_2 = I_{21} + I_{23}.$$

Substitute branch currents:

$$I_2 = y_{12}(V_2 - V_1) + y_{23}(V_2 - V_3).$$

Expand and collect terms:

$$\boxed{I_2 = -y_{12}V_1 + (y_{12} + y_{23})V_2 - y_{23}V_3.}$$

Therefore,

$$Y_{21} = -y_{12}, \quad Y_{22} = y_{12} + y_{23}, \quad Y_{23} = -y_{23}.$$

KCL at Bus 3

At bus 3,

$$I_3 = I_{31} + I_{32}.$$

Substitute branch currents:

$$I_3 = y_{13}(V_3 - V_1) + y_{23}(V_3 - V_2).$$

Expand and collect terms:

$$\boxed{I_3 = -y_{13}V_1 - y_{23}V_2 + (y_{13} + y_{23})V_3.}$$

Therefore,

$$Y_{31} = -y_{13}, \quad Y_{32} = -y_{23}, \quad Y_{33} = y_{13} + y_{23}.$$

Final Y_{bus} for the Three-Bus System

Writing all equations in matrix form,

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} y_{12}+y_{13} & -y_{12} & -y_{13} \\ -y_{12} & y_{12}+y_{23} & -y_{23} \\ -y_{13} & -y_{23} & y_{13}+y_{23} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}.$$



Hence,

$$\mathbf{Y}_{\text{bus}} = \begin{bmatrix} y_{12}+y_{13} & -y_{12} & -y_{13} \\ -y_{12} & y_{12}+y_{23} & -y_{23} \\ -y_{13} & -y_{23} & y_{13}+y_{23} \end{bmatrix}$$

I_1/V_1

$I_1 = (y_{12}+y_{13})V_1$

$I_1 \to V_2$

$-y_{12}$

$I_1 - V_3$

$-y_{13}$

Summary: Complete Bus i Derivation

Branch current:

$$\boxed{I_{ij} = y_{ij}(V_i - V_j)}$$

KCL at bus i:

$$\boxed{I_i = y_i^{sh} V_i + \sum_{j \in \mathcal{N}_i} y_{ij}(V_i - V_j)}$$

Final bus-i equation:

$$\boxed{I_i = \left(y_i^{sh} + \sum_{j \in \mathcal{N}_i} y_{ij} \right) V_i - \sum_{j \in \mathcal{N}_i} y_{ij} V_j}$$

Therefore,

$$\boxed{I_i = \sum_{j=1}^{n} Y_{ij} V_j} \quad \Longrightarrow \quad \boxed{\mathbf{I}_{\text{bus}} = \mathbf{Y}_{\text{bus}} \mathbf{V}_{\text{bus}}.}$$

Note :- There is small mistake in one eqn. → Q → Identify that